



ST BENEDICT'S

SUBJECT	CHEMISTRY EXAMINATION
GRADE	11
EXAMINER	Mr Ntombela
MODERATOR	Mr Hilder
DATE	10 November 2025
MARKS	135
DURATION	2 hours

%

LEARNER NAME:

KEY:

D ☐

E ☐

F ☐

TEACHER:

ZN ☐

KH ☐

FMC ☐

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 17 pages and a separate Data Sheet and Periodic Table. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment.
4. Remember to write your name and surname in the space provided above AND circle your teacher's name
5. Use the data and formulae sheet whenever necessary.
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
8. Answers must be expressed in decimal format, not left as proper fractions.
9. Where appropriate, express answers to TWO decimal places.
10. It is in your own interest to write legibly and to present your work neatly.
11. A Blank page is included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

Question	1	2	3	4	5	6	Total
Mark							
Marker Initial							
Moderated Mark							
Question Total	27	23	24	23	19	19	135

Comment:

Learner's Signature:

Teacher's Signature:

QUESTION 1**27 MARKS**

1.1 Consider the following substances and answer the questions that follow:

A	NH ₃
B	Methane
C	Cu
D	Sodium chloride
E	I ₂
F	Chloromethane

Write down only the LETTER (A – F) to answer the following questions. You can use the same letter more than once, however, only one may be selected for each question.

1.1.1 Is a non-polar molecule? (1)

1.1.2 Is a non-polar molecule with polar covalent bonds? (1)

1.1.3 Is a polar molecule with polar covalent bonds? (1)

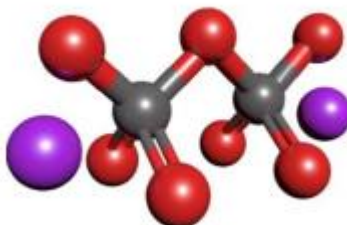
1.1.4 Has hydrogen bond forces. (1)

1.1.5 Forms an ionic crystal lattice structure in solid form. (1)

1.1.6 Exists as a solid in a metallic crystal lattice. (1)

1.1.7 Is a solid at room temperature and is soluble in CCl₄. (1)

1.2 The structure of solid potassium dichromate, K₂Cr₂O₇, is shown in the diagram.



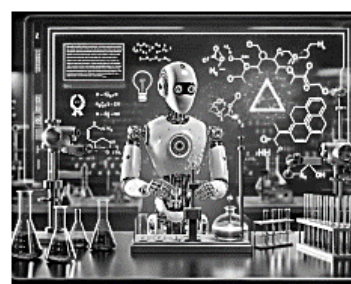
1.2.1 Write the oxidation number for chromium in. Cr₂O₇²⁻. (2)

1.2.2 Name the type/s of bonds that exist in potassium dichromate. (2)

1.2.3 Explain your answer in 1.2.2, with reference to relevant subatomic particles (2)

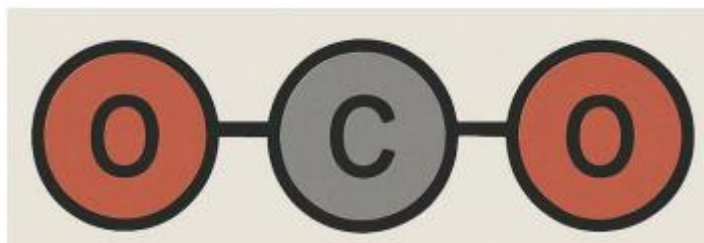
1.3 For better or worse, AI is everywhere, so let's start our exam journey by asking an important question. Can AI replace chemists? Here is a response from Russian chemist Valentine P. Ananikov.

At present, there is little evidence to suggest that AI will replace experimental chemists in research laboratories in the foreseeable future. However, a critical aspect is the competitive advantage that AI confers. Chemists who adeptly integrate AI into their research methodologies are likely to surpass their peers who do not adopt these advanced tools."



*Ananikov V. P., Will artificial intelligence (AI) replace chemists? **Chemistry Today**, 2024, 42(4), 14–15.*

In an attempt to integrate AI into research and be competitive, I asked AI to generate a representation of the carbon dioxide molecule.



1.4 Draw the Lewis diagram of carbon dioxide and hence explain if there are any mistakes present in the AI image above. (2)

1.5 Define the term *electronegativity*. (1)

- 1.6 Use calculations to determine the bond polarity for the C-O bond. (2)

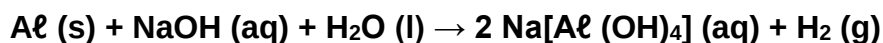
- 1.7 Name the type of intermolecular forces between carbon dioxide molecules and explain how these intermolecular forces arise. (5)

- 1.8 Carbon dioxide has a melting point of $-78,9^{\circ}\text{C}$, methane has a melting point of -182°C . Fully explain these differences in melting point. (4)

QUESTION 2**23 MARKS**

Aluminium plays an important role in the simulated world known as the Matrix where sentient AI machines are enslaving humans. Not only is it a component of the chips that power AI, but aluminium wiring is used to aid in transporting the vast amounts of energy the machines harvest from the human bodies they grow in pods.

Neo decides that he's going to try a different tactic this time and wants to destroy the aluminium infrastructure of the machine world by reacting it with sodium hydroxide which produces sodium aluminate ($\text{Na}[\text{Al}(\text{OH})_4]$) according to the reaction below:



- 2.1 Define the term *reaction rate* for a chemical reaction. (1)

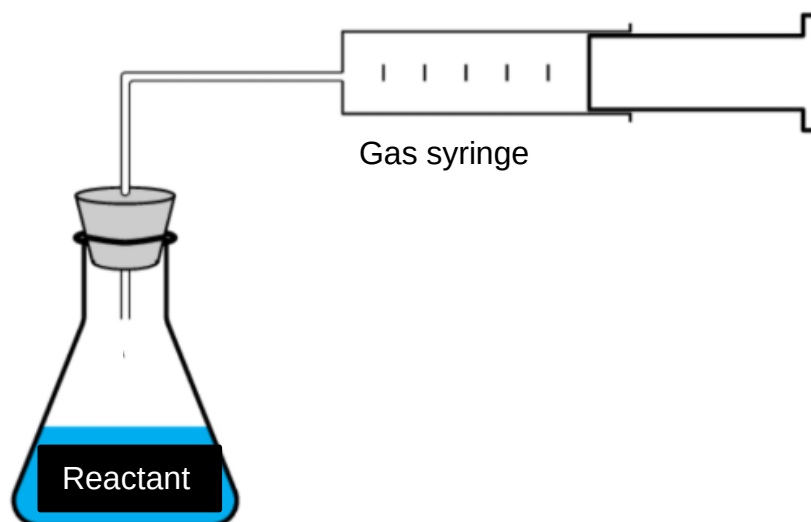
- 2.2 Balance the given partially-balanced equation. (2)



- 2.3 Is the above reaction **EXOTHERMIC** or **ENDOTHERMIC**? Circle your answer (1)

- 2.4 Explain your answer in 2.3. (1)

- 2.5 Morpheus provides Neo with the apparatus shown below. Explain how Neo would use this to measure the rate of the reaction. (2)



Neo runs the experiment and asks Canva AI to plot the results. This is what was generated:

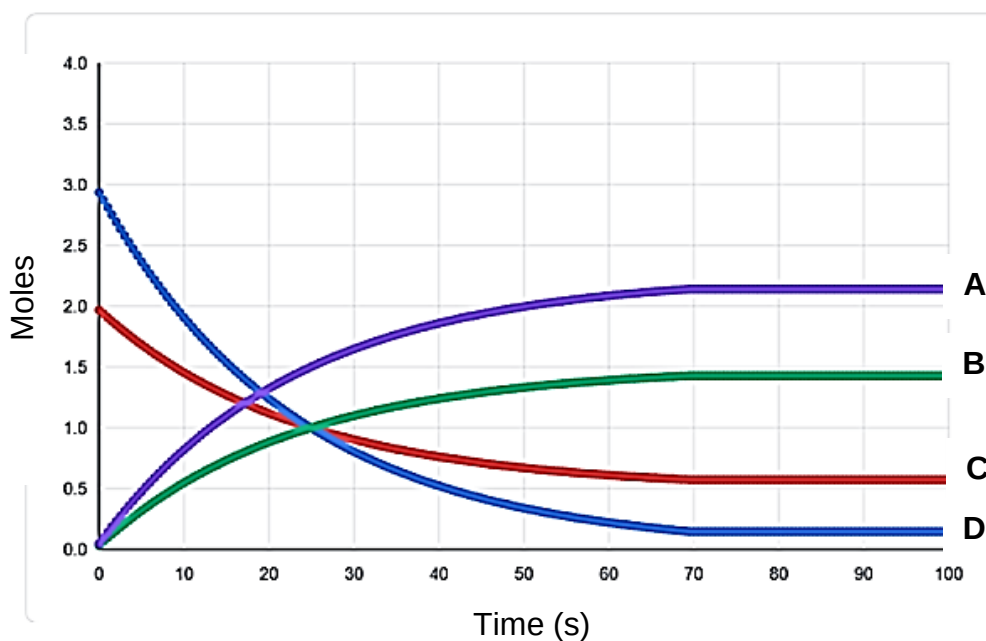
KEY:

A - Hydrogen

B - Sodium
Aluminate

C - Aluminium

D – Sodium
Hydroxide



- 2.6 Canva AI originally plotted a concentration vs time graph for this reaction showing all species except for water. Explain why that would be incorrect. (2)

- 2.7 Calculate the rate of reaction for aluminium. (2)

- 2.8 Neo decreases the temperature of the reaction and runs it again. Draw a fully-labelled Maxwell-Boltzmann curve on the axes below, clearly showing the reaction at the original temperature as T1 and the reaction at the decreased temperature as T2. (4)



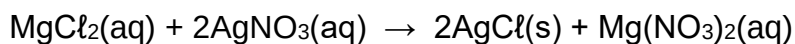
2.9 How does a *catalyst* affect the rate of a reaction? (2)

2.10 Indicate the effect that the addition of a catalyst has on **the graph** on the previous page. (2)

2.11 Use collision theory to explain whether a change in surface area would affect the rate of this reaction. (4)

QUESTION 3**24 MARKS**

- 3.1 Two solutions, one being silver nitrate (AgNO_3) and the other, magnesium chloride (MgCl_2) are mixed together in a glass container. Solid silver chloride (AgCl) forms at the bottom of the container as seen in the following balanced reaction that took place:



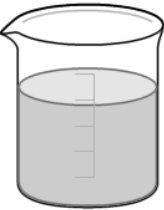
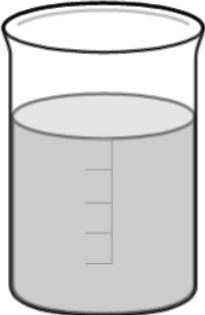
55 cm³ of the magnesium chloride, with a concentration of 0,25 mol·dm⁻³ was mixed with 70 cm³ of silver nitrate solution with a concentration of 0,15 mol·dm⁻³.

- 3.1.1 Prove that the silver nitrate is the limiting reactant. (6)

- 3.1.2 Hence, calculate the mass of silver chloride that can formed during this reaction. (3)

- 3.1.3 Calculate the percentage yield of silver chloride if 1,49 g of silver chloride was in fact recovered. (2)

- 3.2 You are given the following two aqueous solutions:

Solution A $c = 0,2 \text{ mol} \cdot \text{dm}^{-3}$ ammonium hydroxide $V = 0,25 \text{ dm}^3$			Solution B $c = 0,3 \text{ mol} \cdot \text{dm}^{-3}$ ammonium carbonate $V = 0,5 \text{ dm}^3$
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- 3.2.1 Give a reason why the above solutions are *standard solutions*. (1)

- 3.2.2 Define the term *solvent*. (2)

3.2.3 Identify the solvent used in the above solutions. (1)

The balanced dissociation reaction for ammonium hydroxide is given as:



3.2.4 Write balanced chemical equation showing the dissociation of ammonium carbonate. Include phases. (4)

3.2.5 Solution A is now mixed with Solution B in a new bigger beaker.
Calculate the new concentration of the ammonium ions in the solution. (5)

QUESTION 4**23 MARKS**

- 4.1 Hydrogen gas is bubbled through a blue solution of copper (II) sulphate (CuSO_4). No visible reaction occurs. However, when a piece of zinc metal is placed into the same copper (II) sulphate solution, a reddish-brown solid forms on the zinc surface and the blue colour of the solution fades.

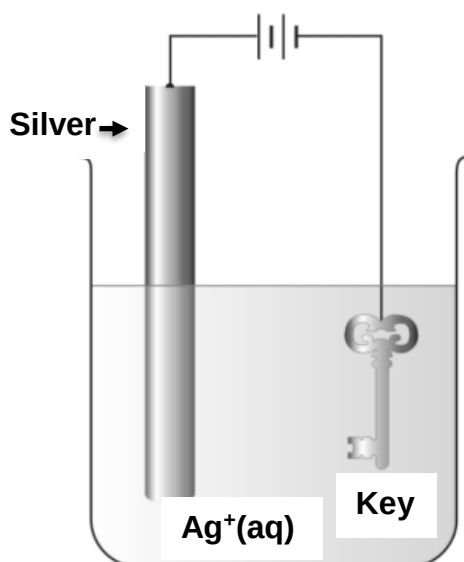
4.1.1 Write the oxidation and reduction half-reactions for the reaction that occurs when zinc metal is placed in copper (II) sulphate solution. (4)

Oxidation:

Reduction:

4.1.2 Use the standard reduction potentials to explain why this reaction is spontaneous, and why the reaction with hydrogen does not occur. (2)

- 4.2 Kwazi now sets up the following electrolytic cell with the aim of coating a key with silver metal, he then dissociated concentrated silver nitrate.



4.2.1 Define *reducing agent*. (1)

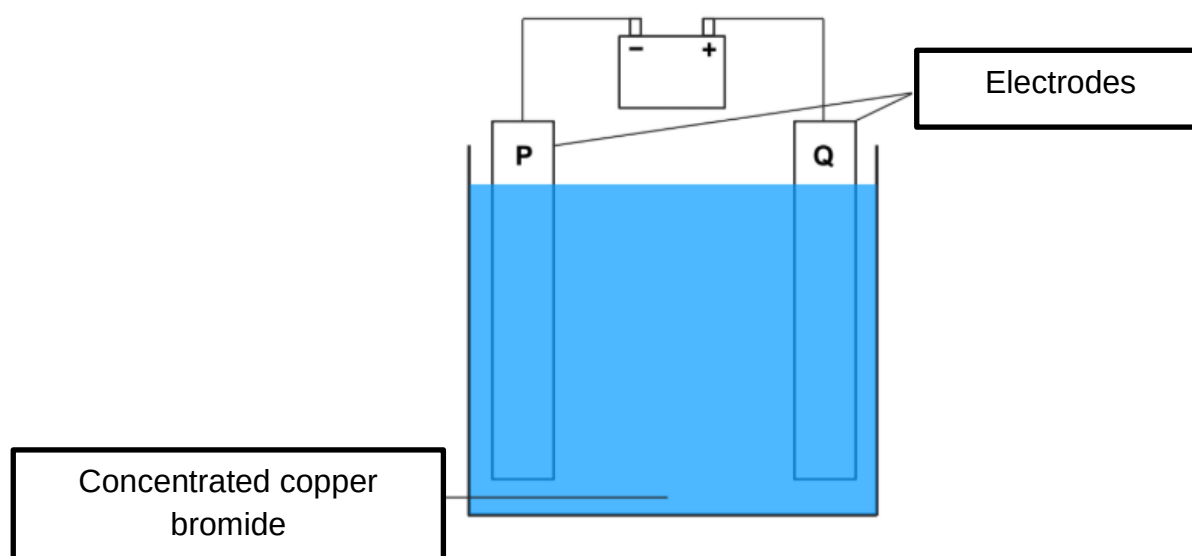
4.2.2 Why is the positive electrode made up of silver? (2)

4.2.3 Show the direction of the silver ion flow in the solution. (1)

4.2.4 Write the equation for the half-reaction taking place at the cathode. (2)

4.2.5 A student says that the key is acting as the reducing agent during electroplating of the key with silver. Do you agree with the student's statement? Give a reason for your answer. (2)

4.3 A CONCENTRATED solution of copper (II) bromide is electrolysed using carbon (graphite) electrodes. The diagram shows a representation of this cell.



Bubbles are observed at the cathode.

4.3.1 Which electrode is the cathode? State only **P** or **Q**. Circle your answer. (1)

4.3.2 Write the equation for the half-reaction taking place at the anode. (2)

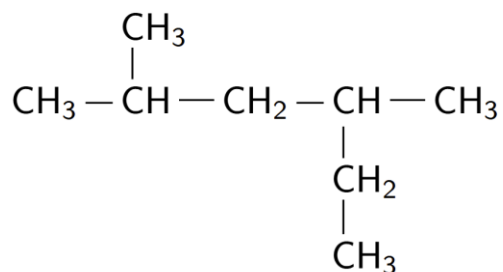
4.3.3 Define *oxidation number*. (1)

- 4.3.4 Use oxidation number to explain which substance is oxidised and which is reduced. (3)

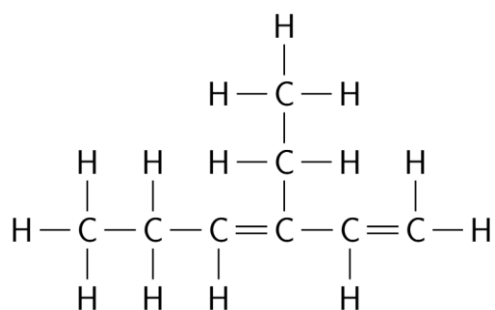
- 4.3.5 Write the balanced net ionic equation for this reaction. (2)

QUESTION 5**19 MARKS**

Consider the two hydrocarbons, compounds **X** and **Y**, shown below.



Compound **X**



Compound **Y**

- 5.1 Define *homologous series*. (2)

- 5.2 Write down the general formula for the homologous series to which compound **X** belongs. (2)

5.3 Classify compound **X** as **SATURATED** or **UNSATURATED**. (2)

5.4 Draw the structural formula for the POSITIONAL ISOMER of compound **X**. (2)

5.5 Write the condensed formula for a chain isomer of compound **X**. (2)

5.6 Describe the functional group of compound **Y**. (1)

5.7 Write the IUPAC name for compound **Y**. (3)

5.8 Which compound is expected to have the **higher boiling point**? (1)

5.9 Explain your answer in 5.8. (4)

QUESTION 6**19 MARKS**

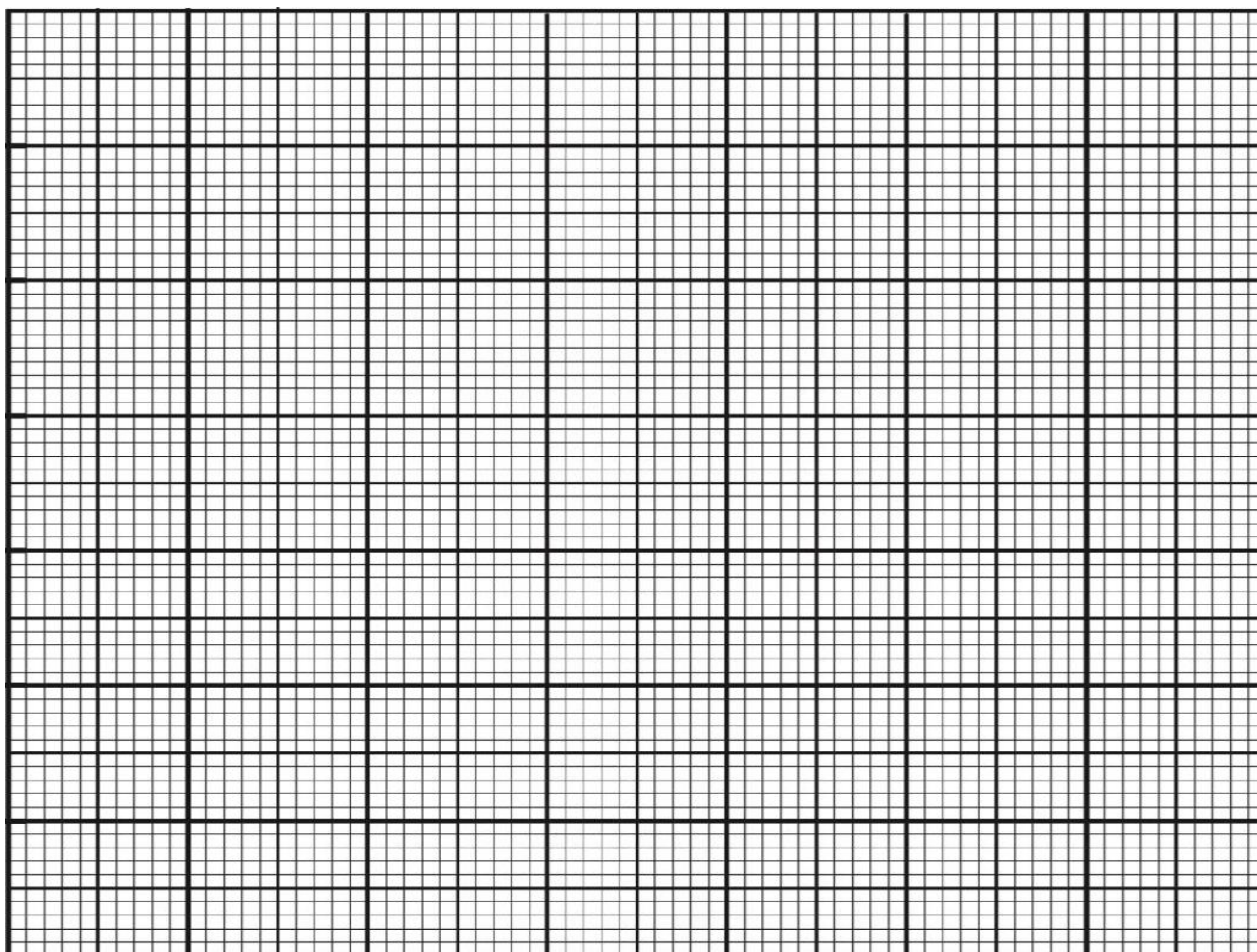
In order to investigate the relationship between the molecular mass of various fluoroalkanes and their boiling point, Jerome collects the data shown in the table below.

Haloalkane	Molecular mass (u)	Boiling point (°C)
CH ₃ F	34	-78
C ₂ H ₅ F	48	-38
C ₄ H ₉ F	76	33
C ₅ H ₁₁ F	90	63

6.1 Identify the predominant intermolecular force in CH₃F. (2)

6.2 Is fluorobutane a GAS or a LIQUID at room temperature (25 °C)? (1)

6.3 Plot a graph of boiling point versus molecular mass for the given data on the axes provided on below. Draw a straight line of best fit. (5)



- 6.4 Jerome wants to use the data he collected to determine the boiling point of $\text{C}_3\text{H}_7\text{F}$. USE THE GRAPH that you drew in Question 6.3 to determine the boiling point of $\text{C}_3\text{H}_7\text{F}$. Indicate clearly on your graph how you arrived at your answer. (2)

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- 6.5 Jerome now wants to use the graph from Question 6.3 to predict the boiling point of chloromethane, CH_3Cl .

6.5.1 Can Jerome use this graph (from Question 6.3) to accurately determine the boiling point of chloromethane? YES/NO. Circle the correct answer. (1)

6.5.2 Briefly explain your answer. (2)

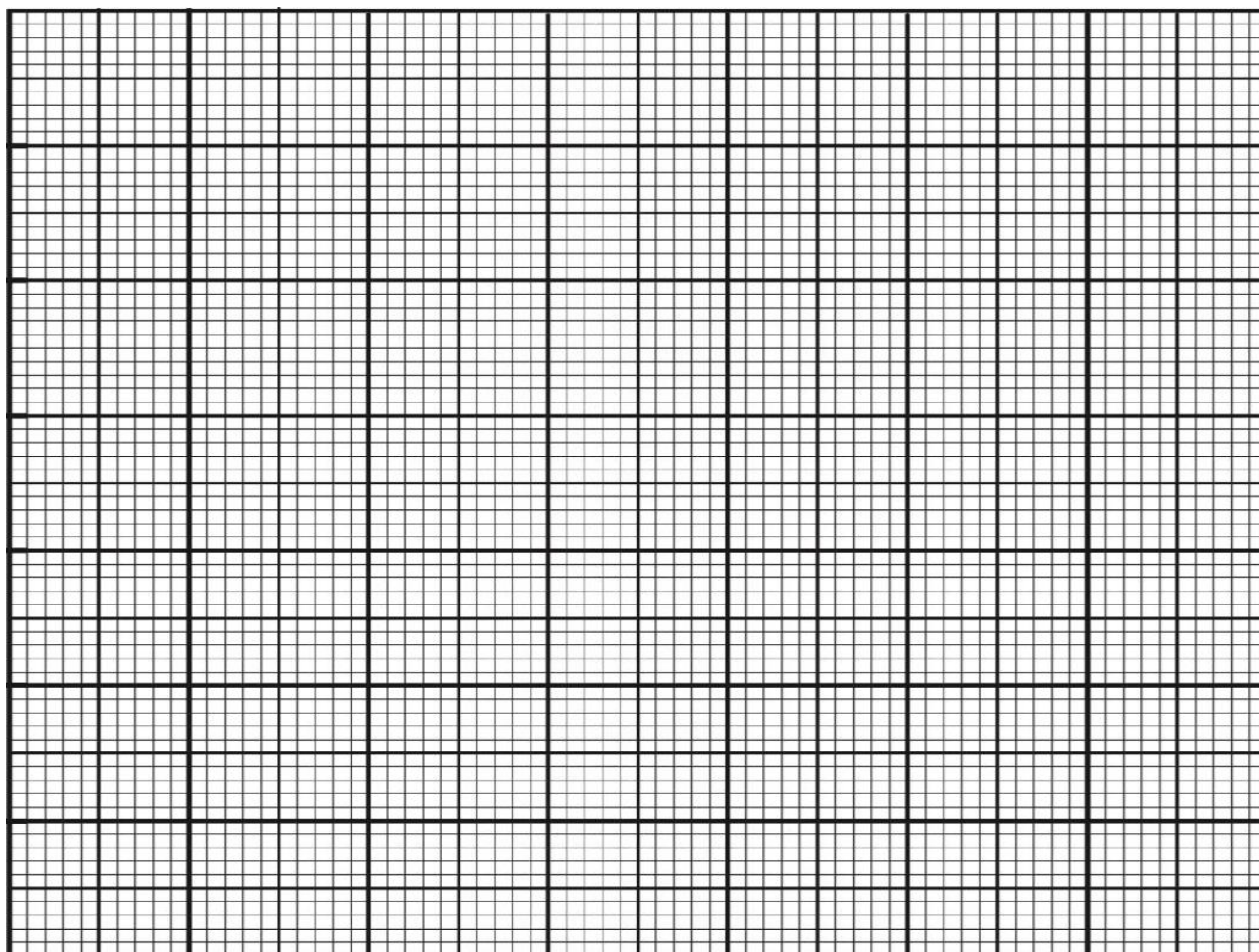
6.6

6.6.1 Define structural *isomers*. (2)

6.6.2 Draw the structural formulae for TWO structural isomers of $\text{C}_3\text{H}_7\text{F}$. (3)

6.6.1 Identify the TYPE of structural isomers you have drawn. (1)

END OF ASSESSMENT

ADDITIONAL GRAPH PAPER

ADDITIONAL SPACE

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